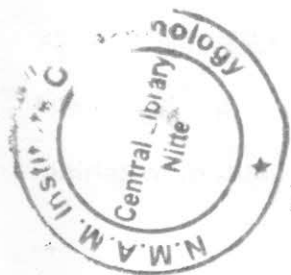


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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fourth Semester B.Tech (CBCS) Degree Examinations

May 2025

CS3004-1 – DESIGN AND ANALYSIS OF ALGORITHMS
 (For CS, IS, AM, CC, AD, CB)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

- How many sub arrays does the quick sort algorithm divide the entire array into?
 A) One B) Two
 C) Three D) Four
- Binary Search works only on which type of arrays?
 A) Unsorted arrays B) Sorted arrays
 C) Arrays with distinct elements D) Arrays of integers
- Linear search algorithm, also known as ____
 A) Binary search B) Simple search
 C) Sequential search D) Difficult search
- In asymptotic analysis, what does it mean if a function is said to be $O(n)$?
 A) The time complexity increases exponentially as n increases B) The algorithm's runtime grows linearly with the input size
 C) The runtime of the algorithm is independent of the input size D) The algorithm's runtime grows logarithmically as n increases
- What will be the output of the following pseudocode if the driver is $\text{fun}(5)$? function $\text{fun}(n)$ if $n == 1$ return 1 else return $n + \text{fun}(n-1)$
 A) 120 B) 24
 C) 6 D) 15
- Pseudocode is written using
 A) Combination of Native & programming language B) High level Programming language only
 C) Natural language only D) Low level programming language
- Which of the following is NOT a characteristic of a good algorithm?
 A) Clarity and simplicity B) Efficiency in terms of time and space
 C) Correctness D) The use of complex data structures
- What is the primary characteristic of the Divide and Conquer approach?
 A) Solving problems by iteration B) Solving problems by dividing them into overlapping sub problems
 C) Dividing a problem into smaller independent subproblems, solving them recursively, and combining their results D) Dividing a problem into smaller problems and linearly solving it.
- Which technique is used to solve the 0/1 Knapsack problem efficiently?
 A) Greedy algorithm B) Divide and conquer
 C) Dynamic programming D) Depth-first search

10. What is the primary purpose of the Floyd-Warshall algorithm?
 A) Finding the shortest path between two specific nodes
 B) Finding the shortest paths between all pairs of nodes in a graph
 C) Finding the minimum spanning tree of a graph
 D) Finding the longest path in a weighted graph
11. In Horspool's algorithm, which table is used to store the shift values for the pattern matching process?
 A) Shift Table
 B) Hash Table
 C) Match Table
 D) Character Table
12. What makes Counting Sort a distribution-based algorithm?
 A) It uses element frequencies to determine sorted positions.
 B) It divides the array into smaller subarrays.
 C) It recursively splits and merges data.
 D) It compares each pair of elements to sort the array.
13. You are given an array: [3, 9, 2, 1, 4, 8]. After building a max-heap from this array, what will be the root element?
 A) 1
 B) 9
 C) 2
 D) 8
14. What operation is performed on an AVL tree when the balance factor of a node becomes greater than 1 or less than -1?
 A) Rotation
 B) Deletion
 C) Insertion
 D) Search
15. Binary Search is a classic example of which variation of Decrease and Conquer?
 A) Decrease by a constant
 B) Decrease by a constant factor
 C) Decrease by one
 D) Divide and conquer
16. In a max-heap, which of the following is true?
 A) The parent node is smaller than its child nodes
 B) The parent node is greater than or equal to its child nodes
 C) The tree is a binary search tree
 D) The elements are stored in ascending order
17. In the Job Assignment Problem, what does the objective function typically aim to minimize?
 A) The time to complete all tasks
 B) The number of tasks assigned
 C) The total cost of assigning tasks to workers
 D) The total time worked by all workers
18. In the subset-sum problem, when do we backtrack?
 A) When the sum of the selected elements exceeds the target sum
 B) When the sum of the selected elements is equal to the target sum
 C) When all elements have been considered
 D) When a subset has been found
19. Dijkstra's Algorithm is the prime example for
 A) Greedy algorithm
 B) Branch and bound
 C) Back tracking
 D) Dynamic programming
20. Which of the following algorithms is the best approach for solving Huffman codes?
 A) exhaustive search
 B) greedy algorithm
 C) brute force algorithm
 D) divide and conquer algorithm

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain the Performance Analysis of an algorithm.	8	L*2	1	1
	b) Write the pseudocode to sort the elements in an array using selection sort and analyze its efficiency. Apply selection sort to sort the following elements in an array. 5, 4, 2, 1, 6, 3.	8	L3	2	2
2.	a) Write the general plan for analyzing the efficiency of non recursive algorithm. Write the pseudocode to find the maximal element in an array and analyze its efficiency.	8	L3	1	2

- b) Write Divide And Conquer recursive Merge sort algorithm and derive the time complexity of this algorithm.
3. a) Write the algorithm for brute force string matching. Show the comparisons the Brute force string matching makes for the pattern $P=acbcac$ in the text $T=acbcabccababcaacbcac$.
- b) Illustrate Quick sort algorithm. Sort the following list using Merge sort. 38, 27, 43, 3, 9, 82, 10.

8	L3	2	2
8	L3	1	2
8	L3	2	2

Unit – II

4. a) Write an insertion sorting algorithm using decrease and conquer technique. Apply the algorithm on the following numbers and trace the steps - 23, 1, 10, 5, 2
- b) Construct AVL tree for the following input by showing each step of the insertion process and type of rotation used for transformation. 63, 9, 19, 27, 18, 108, 99, 81
5. a) Identify the pattern "BARBER" in the given text "JIM_SAW_ME_IN_A_BARBERSHOP" using Horspool's string-matching algorithm. Show the construction of shift table for the same.
- b) Write an algorithm for sorting list of integers by comparison method that uses input enhancement design technique. Apply the algorithm on the following numbers and trace the steps 25, 45, 10, 20, 50, 15.
6. a) Define topological sorting. Apply DFS based method to topologically sort the vertices of the following graph.

8	L3	3	2
8	L3	4	2
8	L3	3	2
8	L3	4	2

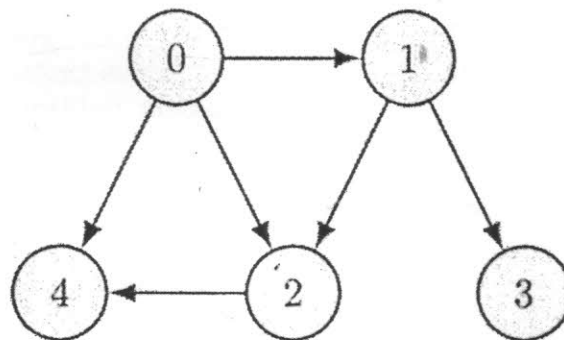


Fig. 6a

- b) Write and Apply Floyd's all pairs shortest path algorithm for the graph shown in the following figure.

8	L3	3	2
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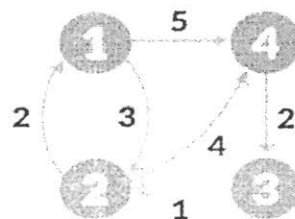


Fig. 6b

8	L3	4	2
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Unit – III

7. a) Solve the given instance of sum of subset problem $S=\{5, 10, 12, 13, 15, 18\}$ and $d=30$. Construct a state space tree.
- b) Define minimum cost spanning tree. Write Prim's algorithm to find minimum cost spanning tree.

8	L3	5	2
8	L2	5	1

8. a) Explain the following: (i) Class P (ii) Class NP (iii) NP Complete Problem (iv) NP Hard Problem.
- b) Solve the job assignment problem using Branch and Bound technique for the following matrix.

8 L2 5 1

	Job 1	Job 2	Job 3	Job 4
A	9	2	7	8
B	6	4	3	7
C	5	8	1	8
D	7	6	9	4

Fig. 8b

8 L3 5 2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.